# Crib Sentry

The baby monitor that alerts instead of only recording.

**Tier 2**

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github.com/SalvarezJ/Crib-Sentry

# The problem

A traditional baby monitor is only useful when someone is watching it. Parents cannot watch all the time and could miss much of what the camera records.

The goal is to cut down on what they miss.

**What the system reports**

- **All clear**
- **Left the crib**
- **Not visible**
- **Hazard present**

### Who cares

Parents and caregivers of infants and toddlers.

### Why it matters

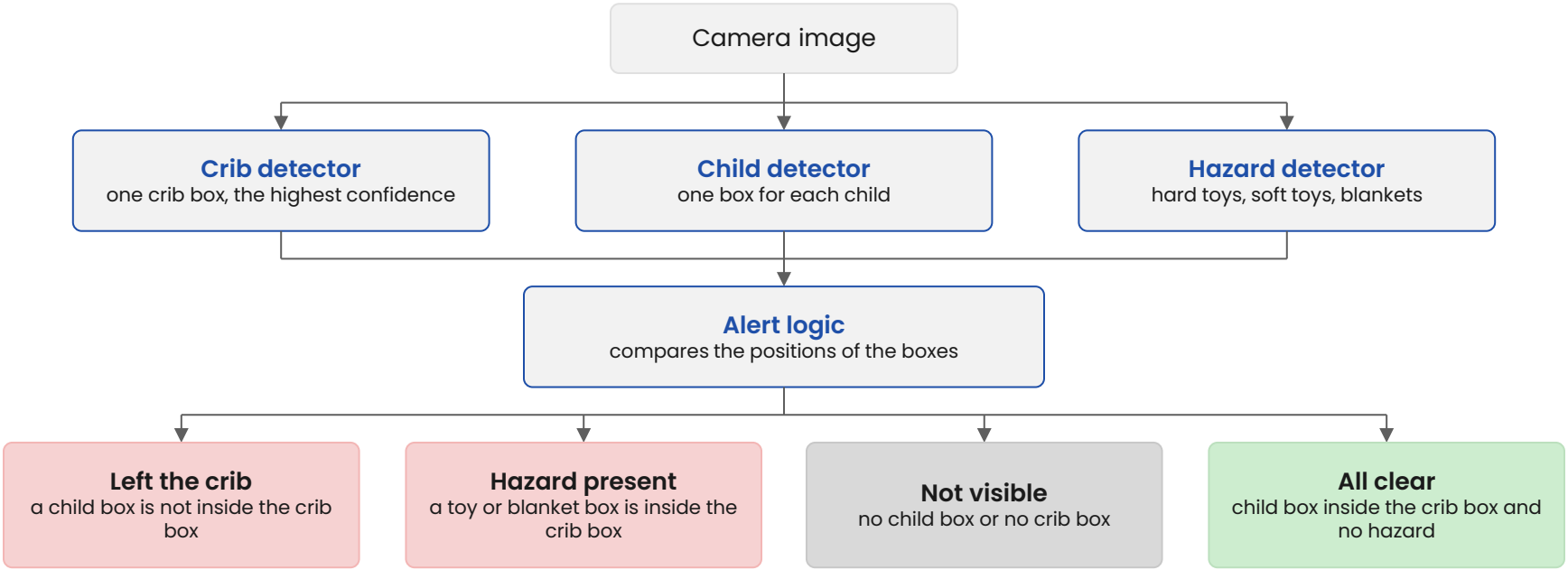
A camera that no person watches gives no protection.

### The solution

A system that looks at the image and gives an alert when one of its conditions is true.

# What I built

One image from a crib camera goes in, three detectors run, and my alert logic compares the boxes and gives one of four alerts.

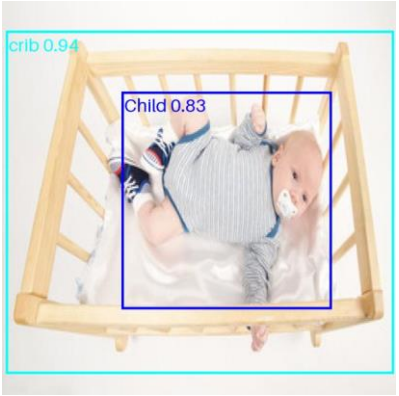



The order of the alerts is from left to right. With no crib box the alert is always not visible.

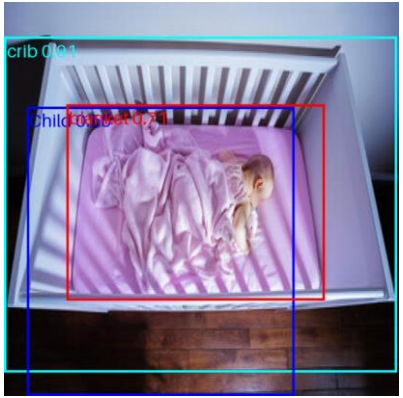
My Blueprint has two detectors. The system now has three.

# Demo

Four correct alerts from my demo notebook. Cyan is the crib box, blue is the child box and red is a hazard box.



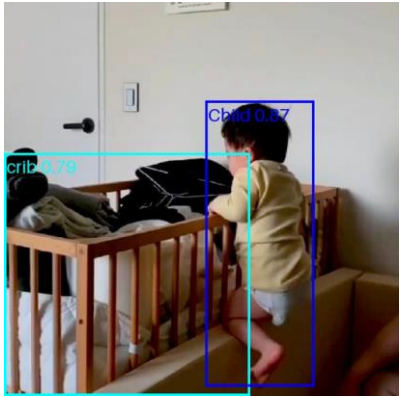

**All clear**
crib 0.94 · child 0.83




**Hazard present**
crib 0.91 · blanket 0.71




**Not visible**
crib 0.87 · no child box




**Left the crib**
crib 0.79 · child 0.87

**Demo video:** https://drive.google.com/file/d/19QUVum-s4lUp3VnR6uf5821H4aUoF2su/view
**Notebook:** notebooks/09_demo.ipynb in my repository

# How it works

CV TECHNIQUE

## Object detection

Each detector draws boxes and gives a confidence number from 0 to 1.

ARCHITECTURE AND MODEL

## CNN: YOLO11n, three models

One finds the crib, one finds the child and one finds toys and blankets.

HOW I USE IT

## Transfer learning

Each model starts from COCO weights and trains for 50 epochs on my images.

FRAMEWORK

## PyTorch through Ultralytics

Free and open source.

Training runs on a free Colab T4 GPU.

**The alert logic.** fraction_inside divides the overlap of two boxes by the area of the inner box. It gives 1 when the box is fully inside and 0 when the boxes do not touch. A box is inside the crib when the result is more than 0.5.

During training I also add darker, lighter, blurred and rotated copies of the images so that the models do better on real rooms.

# The data I used

| Source | Used for | Size | Labels |
|---|---|---|---|
| Baby in Crib | Crib and Child training, and my 24 test images | 242 images (169 / 49 / 24) | Child, Crib |
| Crib_detection | Crib training | 1,494 images | crib, as outlines |
| BASE crib only BABY only | Child training | 309 original images | baby |
| Baby object detection final | Child training, and my 10 "left the crib" test images | 3,136 original frames | baby |
| CribHD T and B | Hazard training | 1,369 training images | hard-toy, soft-toy, blanket |
| CribHD-C | Test only | 120 images | None. I tag each image myself. |

### Cleaning and labels

I remove 18 wrong crib labels and I change outlines into boxes. I keep one file for each original image and I move 101 copies to the training folder.

### Test images stay separate

No detector trains on a test image. I remove five videos from the Child training data because they show or can show my test rooms.

Roboflow datasets: CC BY 4.0. CribHD: non-commercial. Cost: $0.

# Results vs my targets

ALERT ACCURACY · TARGET NOT MET

**70.8%**

17 of 24 test images. My target is 85% or more.

TIME FOR EACH IMAGE · TARGET MET

**0.034 s**

on a T4 GPU. My target is less than 1 second.

| Metric | My Blueprint target | What I achieved |
|---|---|---|
| Alert accuracy on the 24 test images | 85% or more | 70.8% (17 of 24) |
| Time for each image on a T4 GPU | Less than 1 second | 0.034 seconds |
| "Left the crib" on 10 images from a different dataset | No target | 7 of 10. The first system gets 0. |
| Alert accuracy on the 120 CribHD-C images | No target | 41.7% (50 of 120). The first system gets 16.7%. |

**Why CribHD-C is low.** The first system is my Blueprint design, with one detector for the child and the crib. It finds the crib in 22 of 120 images. The new Crib detector finds it in 76, and that takes the accuracy from 16.7% to 41.7%. The child is found in 104, so the missed crib is the main cause and not the blanket.

# What changed from the Blueprint

| My Blueprint | What I build | Why |
|---|---|---|
| Two detectors: one for the child and the crib, one for hazards | Three detectors: Crib, Child and Hazard | The first detector fails on real rooms: 0 of 10 on "left the crib" |
| Two datasets: Baby in Crib and CribHD | Three more datasets for crib and child training | Baby in Crib contains only studio photos |
| Labels used as they are | 18 wrong crib labels removed | They put a crib box on a baby with no crib |
| Risk 2 has a Plan B: find a small extra dataset | 10 "left the crib" test images, and five videos removed from training | No first dataset has a child out of a crib |

**What stays the same**

Object detection, YOLO11, transfer learning, PyTorch through Ultralytics, the four alerts, the two metrics with their targets, and a cost of $0.

# Challenges and fixes

**1. The first system gets 0 of 10 on "left the crib"**

Solution: I train a Crib detector and a Child detector on real rooms, and the system gets 7 of 10. The cost is one alert on my 24 test images; 17 are correct now and 18 before.

**2. A side view misleads my "inside" rule**

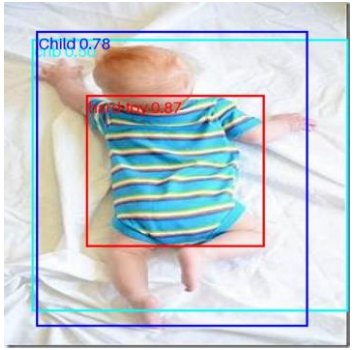
The child climbs out, but more than half of the child box is still on the crib box. Not fixed: the rule works when the camera is above the crib.

**3. The Crib detector is the weakest part**

It puts a weak crib box on photos with no crib, and it finds a crib in only 76 of 120 CribHD-C images. Not fixed: this would be my next step.




**Side view**
Alert: all clear
Correct: left the crib




**No crib in the photo**
Alert: hazard present
Correct: not visible

**Other errors on the 24 test images.** The Hazard detector reads two fitted sheets as blankets. The Child detector does not find one baby.

# Lessons and next steps

MY GREATEST LESSON

**My Blueprint never expected the risk that the system does not find the crib.**

**104**

of 120 CribHD-C images: child found

**76**

of 120 CribHD-C images: crib found

Three of my four alerts need a crib box, so each missed crib is a wrong alert.

**What I build next**

**1. Judge the safety of the child with no crib box**

With no crib box the system gives "not visible", also when it finds a hazard. That works against my own order of alerts.

**2. A minimum confidence for the crib box**

I select it on validation images so that a weak crib box does not start the other rules.

Repository: github.com/SalvarezJ/Crib-Sentry